

TrES-3b

R-2741

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_130620 · created 2026-09-02T01:23:41+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456463.90184 +/- 0.00093 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.141 +/- 0.016
Transit depth (Rp/Rs)^2	1.99 +/- 0.46 [%]
Orbital Inclination (inc)	83.81 +/- 1.34
Airmass coefficient 1 (a1)	1.01 +/- 0.013
Airmass coefficient 2 (a2)	-0.009 +/- 0.012
Scatter in the residuals of the lightcurve fit is	1.26 %
Best Comparison Star	#2 - [568, 157]
Optimal Aperture	3.42
Optimal Annulus	7.59
Transit Duration (day)	0.0658 +/- 0.0092

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.141 $\pm$ 0.016 vs 0.16309 (deviation 1.07 σ)
Duration fit vs published	0.0658 d vs 0.0567 d (deviation 0.77 σ)
Mid-time O-C	1.9 min
Depth SNR	6.9
Residual scatter	1.26 %

Light curve

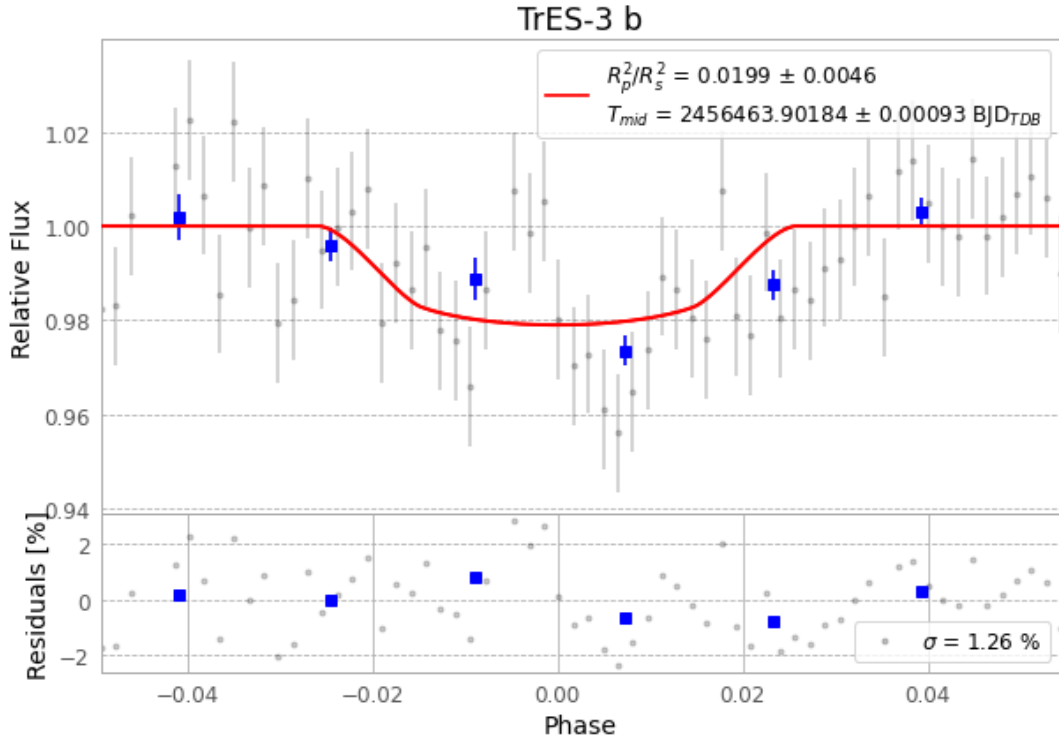


Figure 1 — FinalLightCurve_TrES-3 b_2013-06-20.png (SHA-256 979adb7a7b0c4556...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [363, 267], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 78f67b6b2a0b67cb...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

66 FITS frames (43 MB), TRES-3130620080013.FITS → TRES-3130620111512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-130620.log (554f9b430a96...), FinalLightCurve_TrES-3 b_2013-06-20.png (979adb7a7b0c...), FinalLightCurve_TrES-3 b_2013-06-20.csv (e90d8af7d900...)

Compute resources

Wall time 116.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-09-02 01:23Z.